

第二十九讲：MVCC 和事务隔离级别的设计与实现

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1 UndoLog 和 roll_ptr

2 事务隔离级别

3 实现代码分析

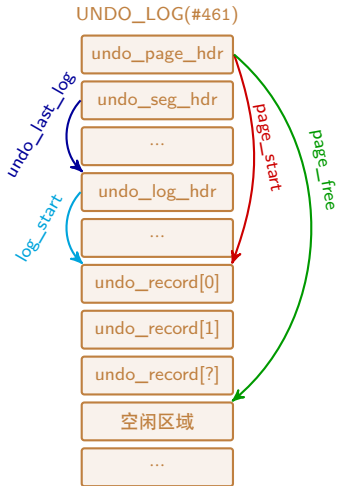
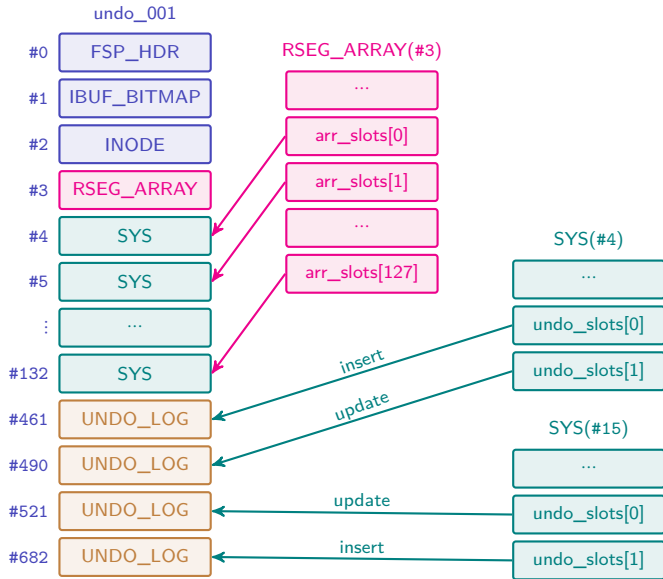


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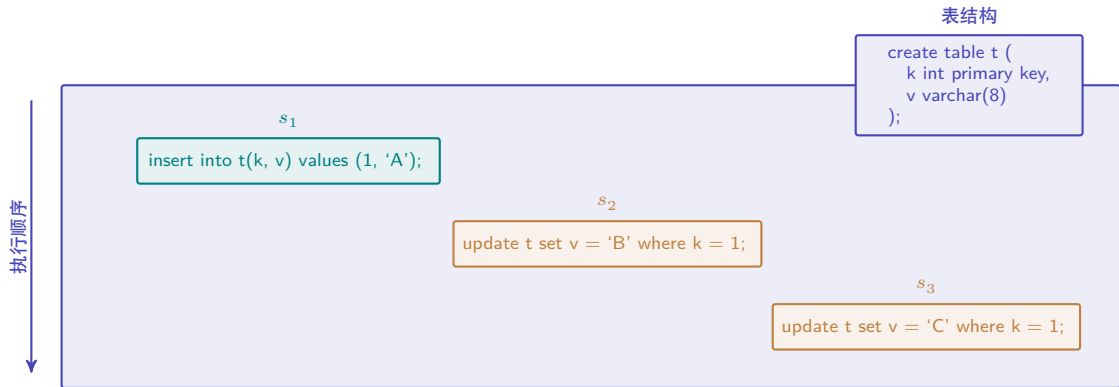
UndoLog 和 roll_ptr



回顾 undo_001 物理结构



单记录插入和更新场景



roll_ptr 回滚段指针

- `trx_undo_report_row_operation()` 函数,

📄 .../trx/trx0rec.cc

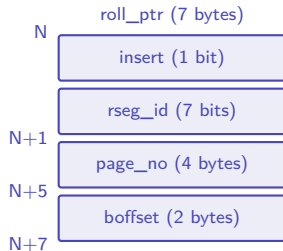
```
2319 *roll_ptr =  
2320     trx_undo_build_roll_ptr(op_type == TRX_UNDO_INSERT_OP,  
2321                             undo_ptr->rseg->space_id, page_no, offset);
```

- ▶ insert 标记是否插入 (Insert) 还是更新 (Update)
- ▶ rseg_id Undo 表空间 ID
- ▶ page_no 所在页码
- ▶ boffset 页内偏移

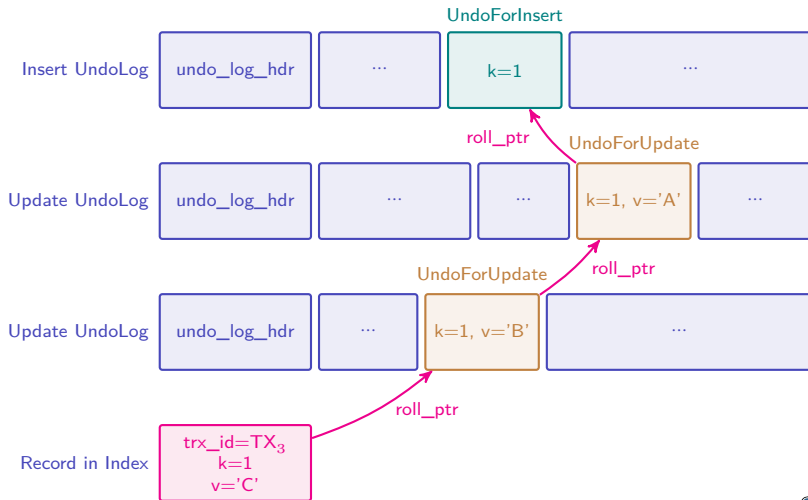
- rseg_id 解析函数,

📄 storage/innobase/include/trx0purge.h

```
229 inline space_id_t id2num(space_id_t space_id) {  
230     if (!is_reserved(space_id)) {  
231         return (space_id);  
232     }  
233  
234     return (((dict_sys_t::s_max_undo_space_id - space_id) %  
235             FSP_MAX_UNDO_TABLESPACES) +  
236             1);  
237 }
```

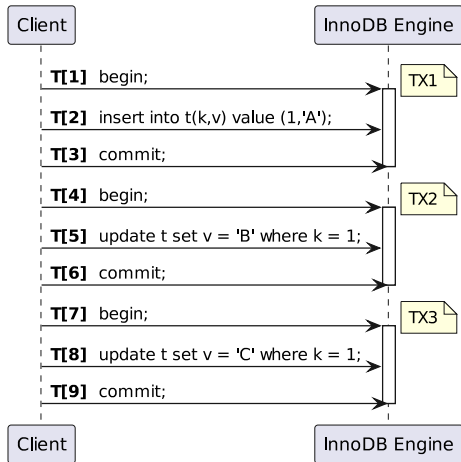


UndoLog 构成的版本链

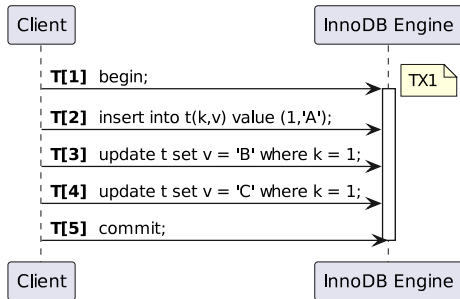


事务对 Undo 版本链影响的场景

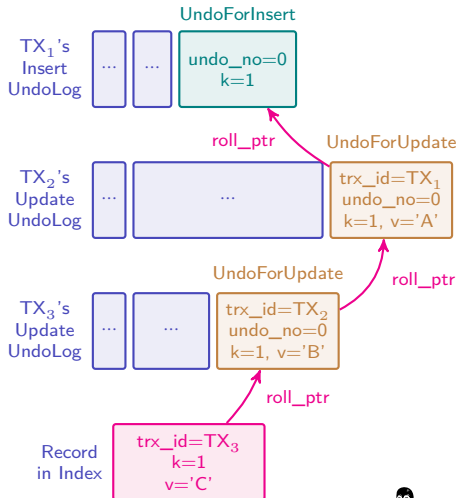
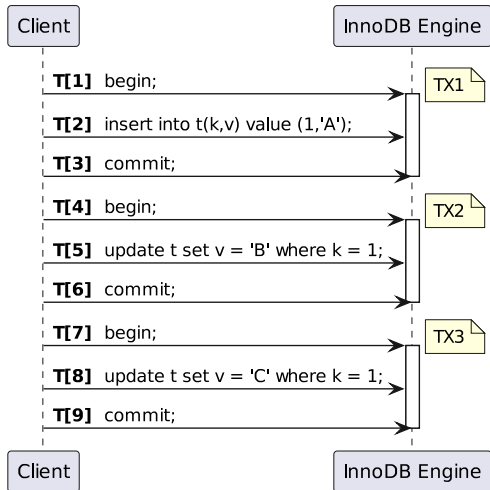
场景一：每个语句位于不同的事务内



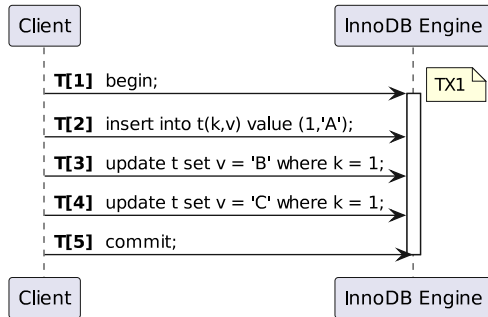
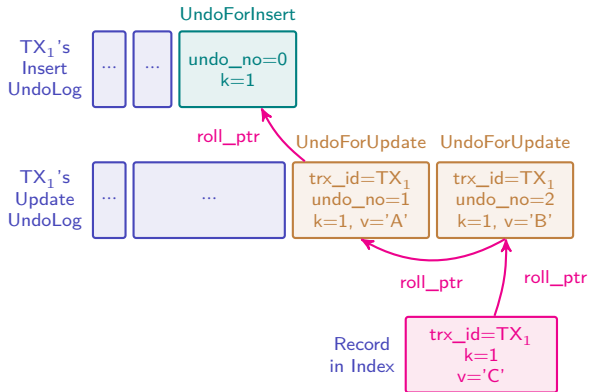
场景二：所有语句位于相同的事务内



场景一：不同事务产生的 Undo 版本链



场景二：相同事务产生的 Undo 版本链



2

事务隔离级别



TX₁ 和 TX₂ 执行过程分析

执行顺序
↓

表结构

```
create table t (  
  k int primary key,  
  v varchar(8)  
);
```

TX₁

delete from t;

s₀ insert into t(k, v) values (1, 'A');

begin;

s₀

k=1, v=A

s₁

k=1, v=?

s₂

k=1, v=?

s₁ select * from t;

s₂ select * from t;

TX₂

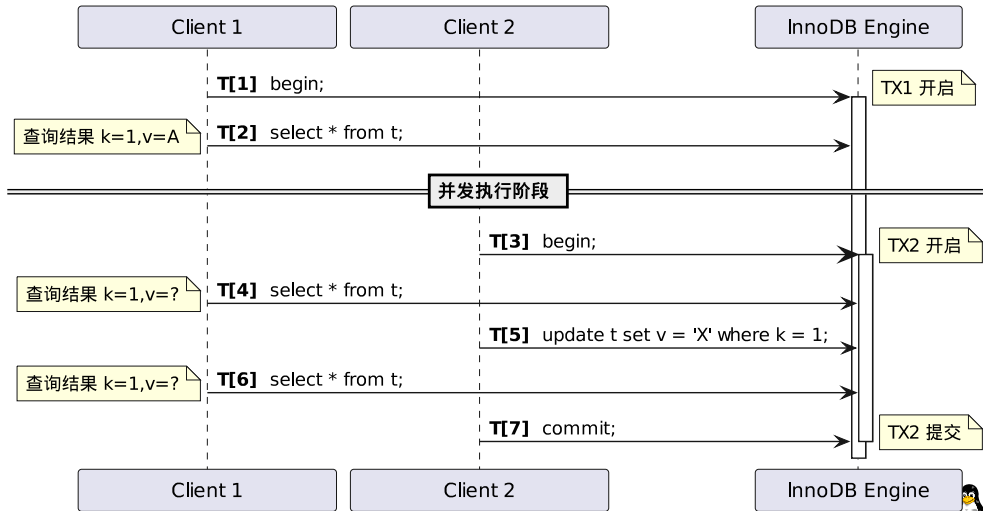
begin;

update t set v='X' where k=1;

commit;



多事务执行时序图



RR 可重复读

----- INIT -----	----- TX1 -----	----- TX2 -----
mysql> show variables like '%iso%\G	mysql> begin;	mysql>
***** 1. row *****	mysql>	mysql> begin;
Variable_name: transaction_isolation	mysql>	mysql> update t set v = 'X' where k = 1;
Value: REPEATABLE-READ	mysql>	Query OK, 1 row affected (0.00 sec)
1 row in set (0.02 sec)	mysql>	Rows matched: 1 Changed: 1 Warnings: 0
	mysql> select * from t;	
mysql> select * from t;	+---+-----+	mysql>
+---+-----+	k v	mysql>
k v	+---+-----+	mysql>
+---+-----+	1 A	mysql>
1 A	+---+-----+	mysql>
+---+-----+	1 row in set (0.00 sec)	mysql> commit;
1 row in set (0.00 sec)		Query OK, 0 rows affected (0.00 sec)
	mysql>	
mysql>	mysql> select * from t;	mysql>
	+---+-----+	mysql>
	k v	mysql>
	+---+-----+	mysql>
	1 A	mysql>
	+---+-----+	mysql>
	1 row in set (0.00 sec)	mysql>
		mysql>
	mysql> commit;	mysql>
	Query OK, 0 rows affected (0.00 sec)	mysql>



RC 读已提交

----- INIT -----	----- TX1 -----	----- TX2 -----
mysql> show variables like '%iso%\G	mysql> begin;	mysql>
***** 1. row *****	mysql>	mysql> begin;
Variable_name: transaction_isolation	mysql>	mysql> update t set v = 'X' where k = 1;
Value: READ-COMMITTED	mysql>	Query OK, 1 row affected (0.00 sec)
1 row in set (0.02 sec)	mysql>	Rows matched: 1 Changed: 1 Warnings: 0
	mysql> select * from t;	
mysql> select * from t;	+---+-----+	mysql>
+---+-----+	k v	mysql>
k v	+---+-----+	mysql>
+---+-----+	1 A	mysql>
1 A	+---+-----+	mysql>
+---+-----+	1 row in set (0.00 sec)	mysql> commit;
1 row in set (0.00 sec)		Query OK, 0 rows affected (0.00 sec)
	mysql>	
mysql>	mysql> select * from t;	mysql>
	+---+-----+	mysql>
	k v	mysql>
	+---+-----+	mysql>
	1 X	mysql>
	+---+-----+	mysql>
	1 row in set (0.00 sec)	mysql>
		mysql>
	mysql> commit;	mysql>
	Query OK, 0 rows affected (0.00 sec)	mysql>



RU 读未提交

----- INIT -----	----- TX1 -----	----- TX2 -----
mysql> show variables like '%iso%'\G	mysql> begin;	mysql>
***** 1. row *****	mysql>	mysql> begin;
Variable_name: transaction_isolation	mysql>	mysql> update t set v = 'X' where k = 1;
Value: READ-UNCOMMITTED	mysql>	Query OK, 1 row affected (0.00 sec)
1 row in set (0.02 sec)	mysql>	Rows matched: 1 Changed: 1 Warnings: 0
	mysql> select * from t;	
mysql> select * from t;	+---+-----+	mysql>
+---+-----+	k v	mysql>
k v	+---+-----+	mysql>
+---+-----+	1 X	mysql>
1 A	+---+-----+	mysql>
+---+-----+	1 row in set (0.00 sec)	mysql> commit;
1 row in set (0.00 sec)		Query OK, 0 rows affected (0.00 sec)
	mysql>	
mysql>	mysql> select * from t;	mysql>
	+---+-----+	mysql>
	k v	mysql>
	+---+-----+	mysql>
	1 X	mysql>
	+---+-----+	mysql>
	1 row in set (0.00 sec)	mysql>
		mysql>
	mysql> commit;	mysql>
	Query OK, 0 rows affected (0.00 sec)	mysql>



S 串行化

----- INIT -----	----- TX1 -----	----- TX2 -----
mysql> show variables like '%iso%\G ***** 1. row ***** Variable_name: transaction_isolation Value: SERIALIZABLE 1 row in set (0.02 sec)	mysql> begin; mysql> mysql> mysql> mysql> select * from t; 阻塞	mysql> mysql> begin; mysql> update t set v = 'X' where k = 1; Query OK, 1 row affected (0.00 sec) Rows matched: 1 Changed: 1 Warnings: 0 mysql>
mysql> select * from t; +----+-----+ k v +----+-----+ 1 A +----+-----+ 1 row in set (0.00 sec)		
mysql>		

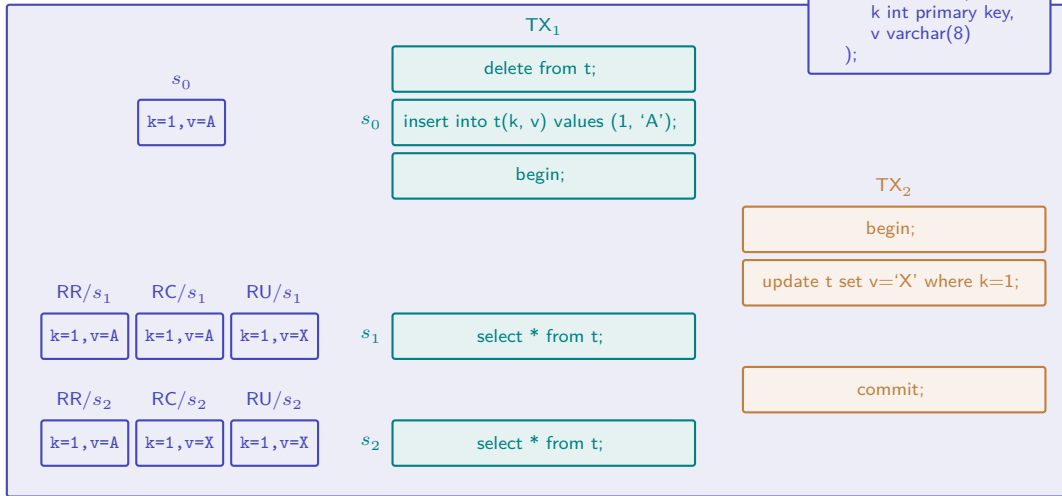


RR/RC/RU 隔离级别汇总

表结构

```
create table t (  
  k int primary key,  
  v varchar(8)  
);
```

执行顺序



3

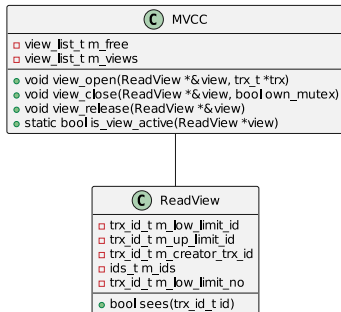
实现代码分析



ReadView

● ReadView 类重要成员变量定义 ... /include/read0types.h

```
282  /** The read should not see any transaction with trx id >= this
283  value. In other words, this is the "high water mark". */
284  trx_id_t m_low_limit_id;
285
286  /** The read should see all trx ids which are strictly
287  smaller (<) than this value. In other words, this is the
288  low water mark". */
289  trx_id_t m_up_limit_id;
290
291  /** trx id of creating transaction, set to TRX_ID_MAX for free
292  views. */
293  trx_id_t m_creator_trx_id;
294
295  /** Set of RW transactions that was active when this snapshot
296  was taken */
297  ids_t m_ids;
298
299  /** The view does not need to see the undo logs for transactions
300  whose transaction number is strictly smaller (<) than this value:
301  they can be removed in purge if not needed by other views */
302  trx_id_t m_low_limit_no;
```



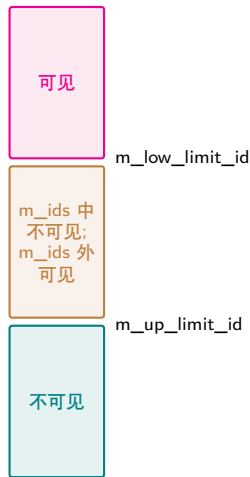
RC 和 RR 下的 ReadView

- RC 和 RR 下生成 ReadView 的时机是有所差异的

- ▶ RC 每次 SELECT 数据前都生成一个 ReadView
- ▶ RR 只在第一次读取数据时生成一个 ReadView, 后面会复用第一次生成的
- ▶ RU 和 S 是两个特殊的隔离级别, 它们不需要使用 ReadView

- 函数调用点

```
gdb) bt 3
#0  MVCC::view_open (this=0x7fffe4323800, view=@0x7fffe95f2098: 0x0, trx=0x7fffe95f1ff8)
    at /opt/src/mysql-server/storage/innobase/read/read0read.cc:530
#1  0x000055555aa1ac98 in trx_assign_read_view (trx=0x7fffe95f1ff8)
    at /opt/src/mysql-server/storage/innobase/trx/trx0trx.cc:2324
#2  0x000055555a952747 in row_search_mvcc (buf=0x7fff38b52b80 "\377", mode=PAGE_CUR_G,
    prebuilt=0x7fff380165a8, match_mode=0, direction=0)
    at /opt/src/mysql-server/storage/innobase/row/row0sel.cc:4822
(More stack frames follow...)
(gdb)
```



结束

